

MSC THESIS DEFENSE

Data-Driven Cognitive Phenotyping in Acquired Brain Injury

Do routine neuropsychological data reveal cognitive subtypes?

Zoltan Kunos

MSc Fundamental Principles of Data Science · Universitat de Barcelona

Supervisor: Dr. Alejandro García-Rudolph

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The clinical problem: heterogeneity

- ABI can disrupt attention, memory, language, executive function, orientation and visuospatial cognition.
- Impairment does not follow the same pattern in every patient.
- Diagnosis and injury severity are useful categories — but they do not capture cognitive heterogeneity.
- Incomplete assessments are common, especially among more impaired patients.

Research question

Can data-driven clustering recover clinically meaningful cognitive organisation when missing data are handled explicitly — rather than ignored?

Research gap and contribution

Common limitation in prior work	This dissertation
Samples often below 200 participants	17,406 assessments · 7,285 patients
Complete-case analysis	Ten imputation strategies compared
Limited algorithmic sensitivity testing	Hyperparameter, seed and missingness checks
Predetermined number of clusters	Density-based clustering (HDBSCAN)
Limited clinical validation	Association with clinical-unit assignment

Four hypotheses, objective criteria

H1

Are stable clusters recoverable?

Clusters separate cleanly, leave few cases unassigned, and form at least two groups

`Silhouette > 0.40 · noise < 30% · ≥ 2 clusters`

H2

Robust to imputation choice?

Cluster assignments agree strongly whichever imputation method is used

`Mean pairwise ARI > 0.50 (fixed-reference)`

H3

Are tiers clinically relevant?

Severity tier relates to clinical unit, even after accounting for diagnosis

`$\chi^2$  test · Cramér's V > 0.10 · CMH adjust for diagnosis`

H4

Does domain aggregation help?

Grouping variables into domains gives cleaner, more stable separation

`Higher silhouette · lower VIF & condition number`

THE DATA

A large but incomplete cohort

17,406

assessments

7,285

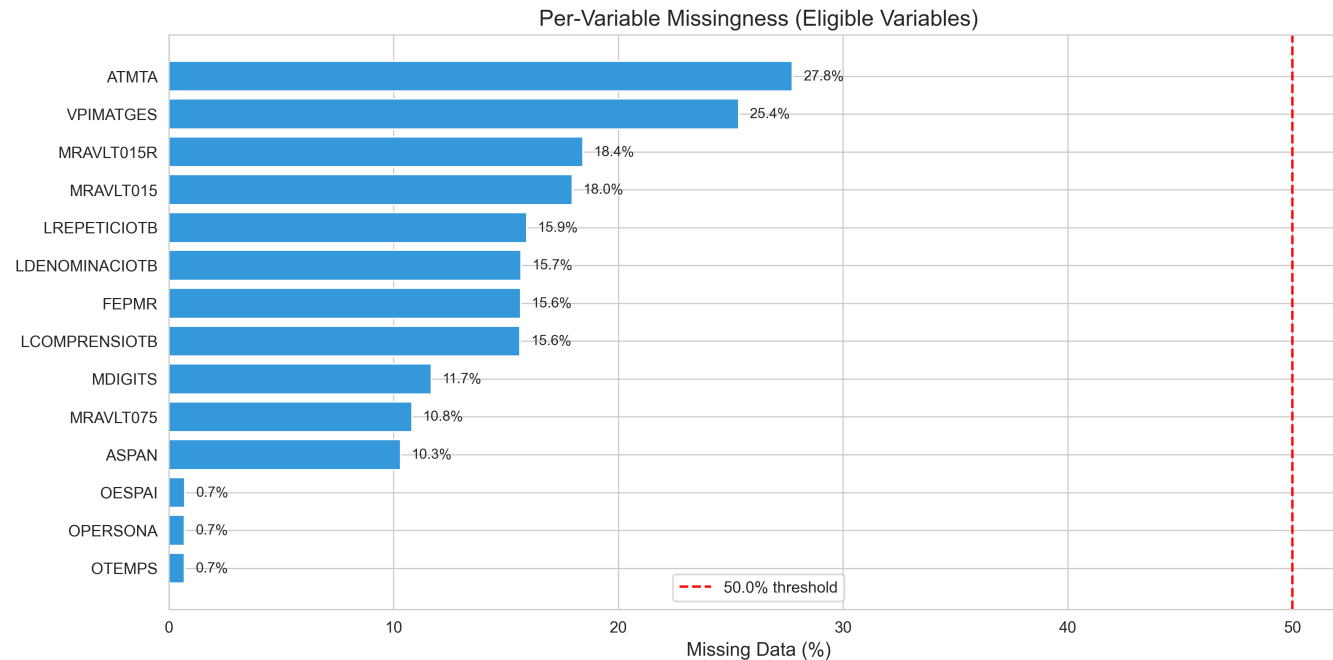
patients

14

variables

6

cognitive domains

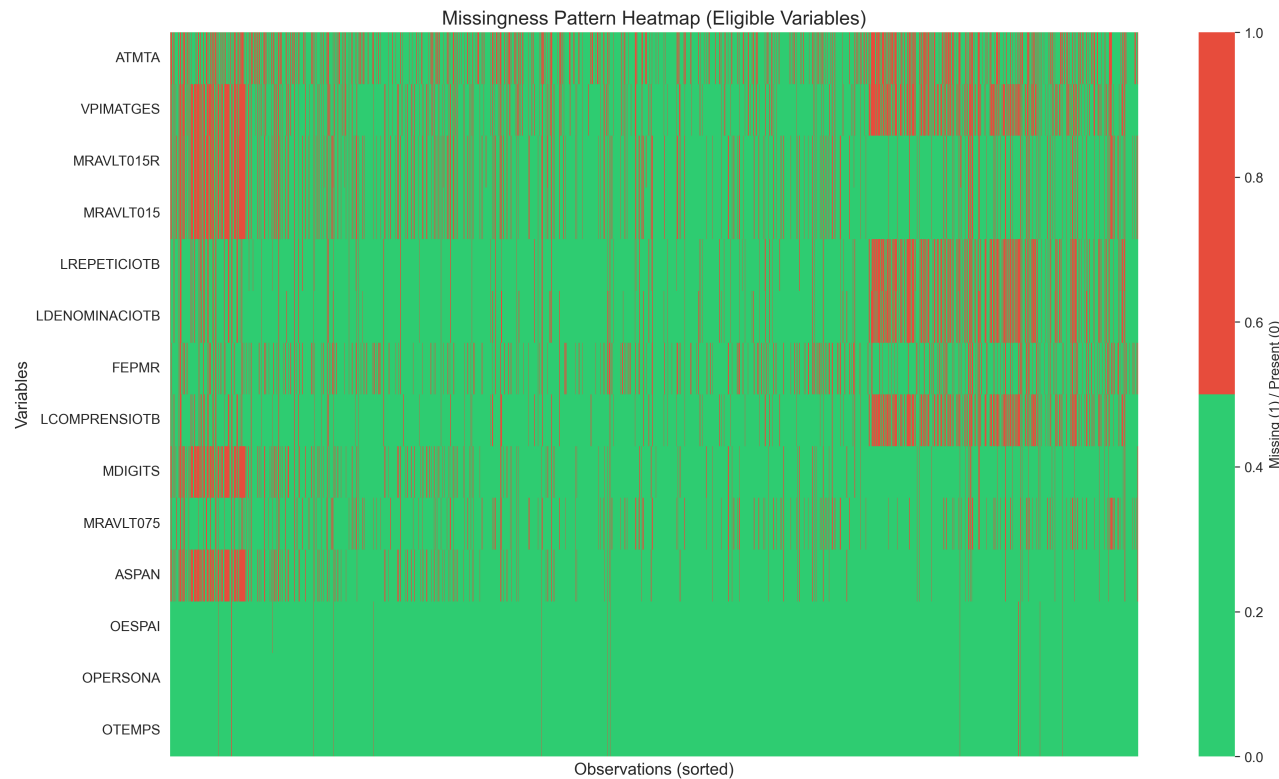


53.1%

of rows are complete across all 14 variables

- Complete-case analysis would discard $\approx$ 47% of assessments.
- Missingness is non-random — it tracks the testing process.
- Imputation is needed to retain partially observed patients.

Why imputation is necessary

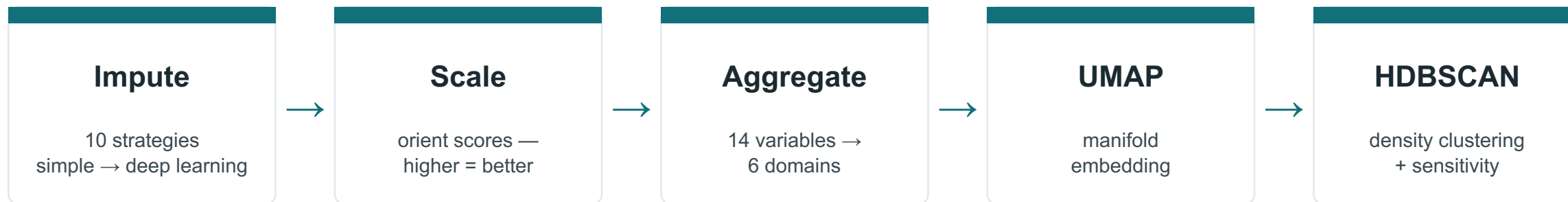


Missingness reflects the clinical testing process

- Fatigue or reduced motivation
- Cognitive overload or task aversion
- Disorientation or motor limitations
- Limited tolerance for long batteries
- Clinician selects a more suitable test

Dropping incomplete rows preferentially removes the most impaired patients — so missingness must be managed and tested, not hidden.

Analysis pipeline



Fixed-reference design (H2)

One MICE solution freezes the scaler, UMAP embedding and HDBSCAN partition; every other imputation is then projected through this fixed structure — its missing values placed on the same map and into the same clusters.

✓ **Tests** the same patient lands in the same tier whichever imputation filled the gaps — geometry held fixed.

✗ **Not tested** that re-fitting the whole pipeline per dataset would rediscover the same cluster hierarchy.

UMAP

Uniform Manifold Approximation & Projection · McInnes, Healy & Melville, 2018

- Builds a fuzzy k-NN graph of neighbourhood structure.
- Balances local and global geometry.
- Reduces dimensionality before density clustering.

Caveat: separation in UMAP space can be optimistic — it must be backed by sensitivity analyses.

FORMAL DEFINITION

$$p_{j|i} = \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right), \quad \sum_j p_{j|i} = \log_2 k$$

$$p_{ij} = p_{j|i} + p_{i|j} - p_{j|i} p_{i|j}$$

$$q_{ij} = \left(1 + a \|y_i - y_j\|_2^{2b}\right)^{-1}$$

$$\mathcal{C} = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}} + (1 - p_{ij}) \log \frac{1 - p_{ij}}{1 - q_{ij}}$$

HDBSCAN

Hierarchical Density-Based Spatial Clustering of Applications with Noise · Campello, Moulavi & Sander, 2013

FORMAL DEFINITION

- Needs no predefined number of clusters.
- Finds regions dense across density levels.
- Labels uncertain points as noise, not forced.

$$\text{core}_k(x) = d(x, N_k(x))$$

$$d_{\text{mreach}}(a, b) = \max\{\text{core}_k(a), \text{core}_k(b), d(a, b)\}$$

$$S(C) = \sum_{x \in C} (\lambda_{\max}(x) - \lambda_{\text{birth}}(C)), \quad \lambda = \frac{1}{\text{dist}}$$

$$\hat{\mathcal{P}} = \arg \max_{\mathcal{P}} \sum_{C \in \mathcal{P}} S(C) \quad (\text{non-nested})$$

The minimum cluster size is the main knob; everything else follows from the density hierarchy.

Ten imputation strategies

Simple → model-based

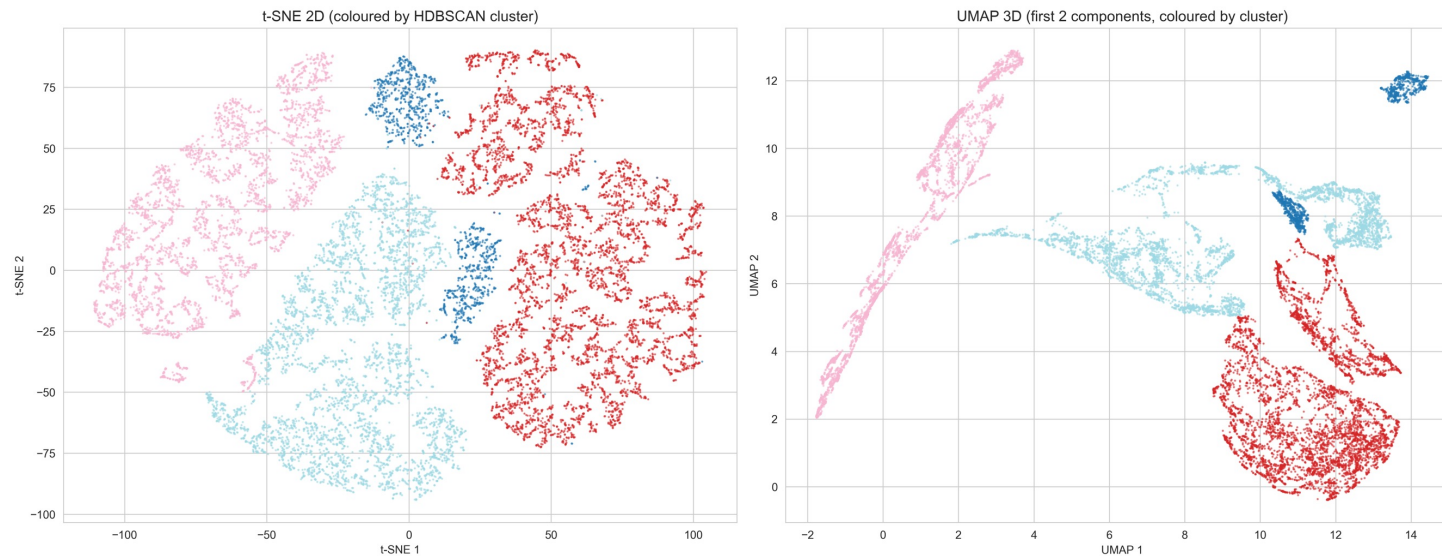
1. Mean imputation
2. Median imputation
3. Predictive mean matching
4. Expectation–maximization
5. MICE (reference)

Donor / matrix / deep learning

6. K-nearest neighbours
7. MissForest
8. SoftImpute
9. Non-negative matrix factorization
10. Deep generative approaches

Agreement across this full range is what makes the recovered tiers credible — not an artefact of one preprocessing choice. Settings in Appendix A.

Three cognitive severity tiers emerge



Above-Average

above cohort mean

6,725

Near-Normal

close to the average

6,328

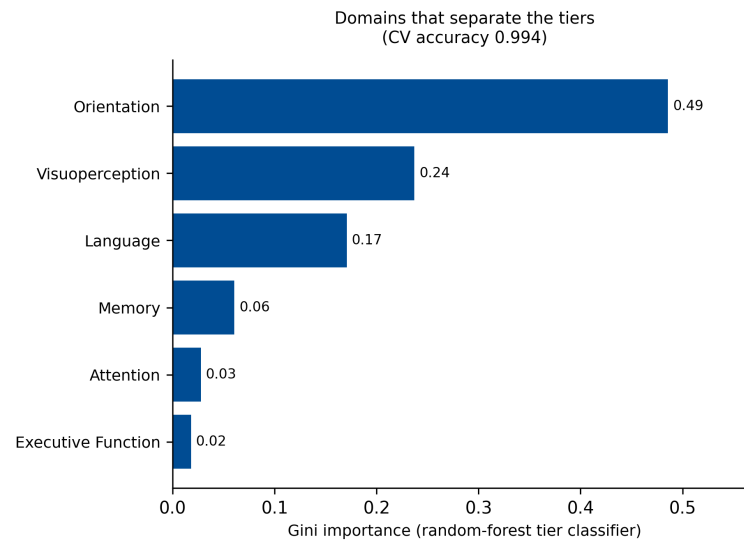
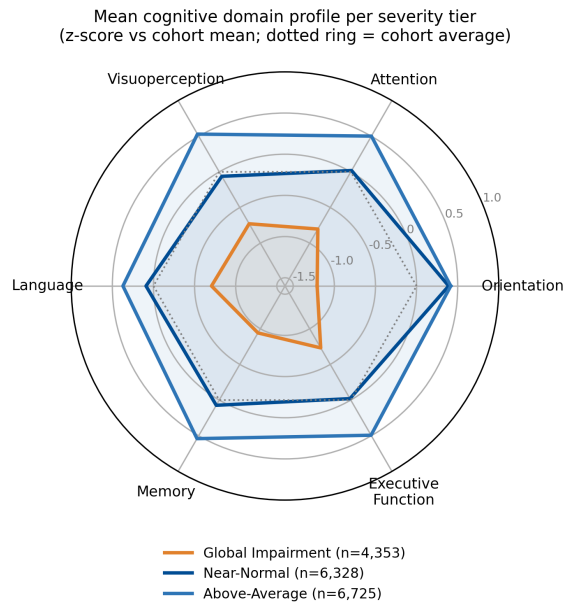
Global Impairment

below cohort mean

4,353

All 10 imputations cleared pre-registered thresholds · MICE reference silhouette ≈ 0.53 · $k = 3$. Criteria met, but the tiers are not interpreted as essential categories.

Continuum evidence: level, not shape



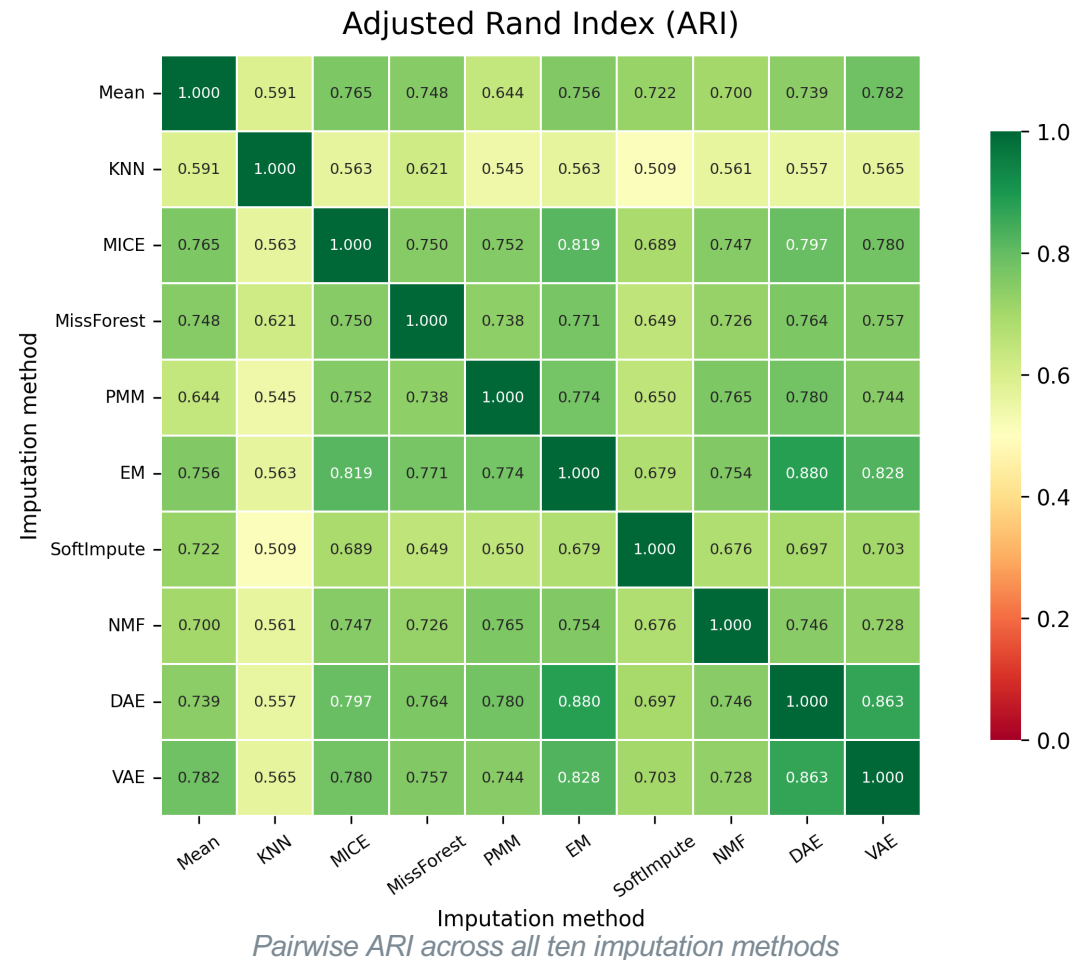
PC1

≈ 56%

of variance — one dominant axis

- Radar profiles are roughly concentric.
- All domains move in the same direction.
- All domains load positively on PC1.
- Signal = global severity, not dissociation.

Robust across imputation methods



0.71

mean pairwise ARI (95% CI 0.70–0.72)

44 / 45

method pairs survive Holm correction

> 86%

assessments with high-confidence consensus

Scope: point assignments are robust under a fixed scaler, embedding and partition — not full invariance of independently re-fit pipelines.

Complete-case analysis changes the answer

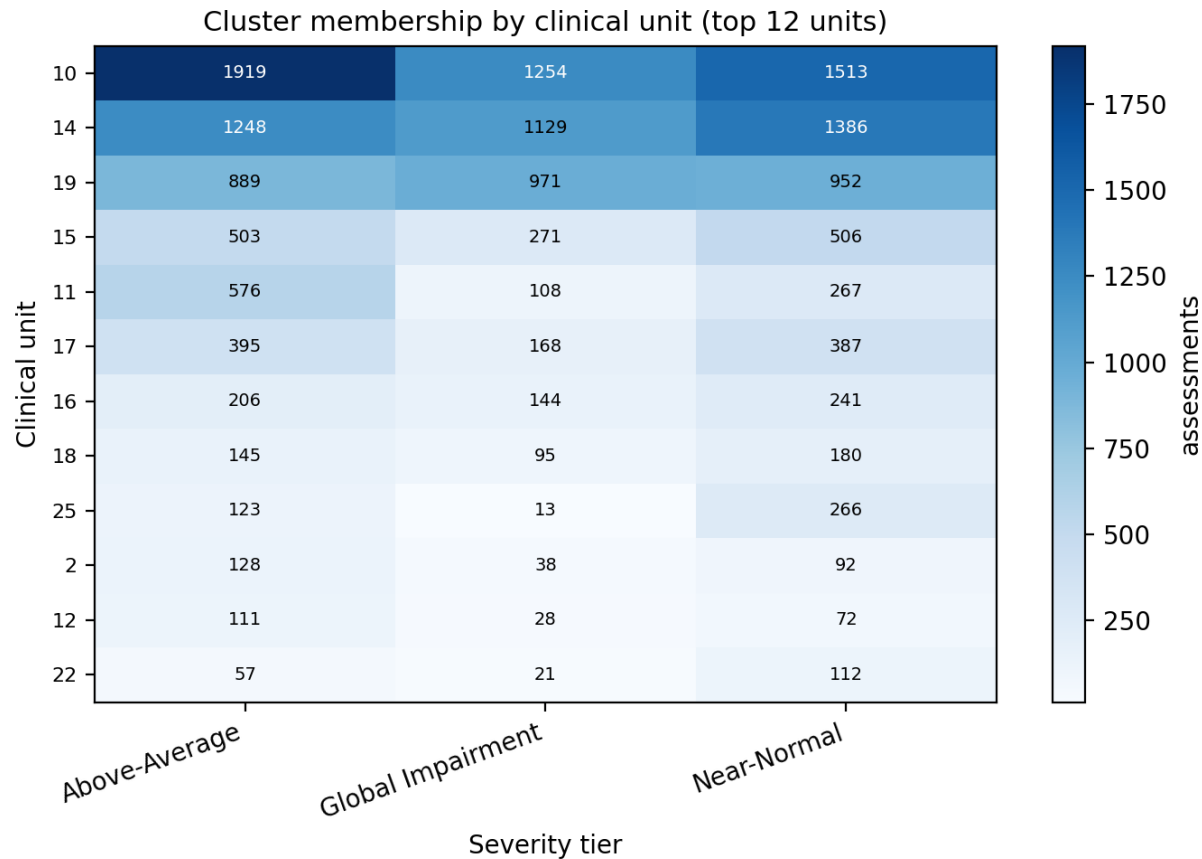
	MICE reference	Complete-case
Assessments retained	100%	53.1%
Tiers recovered	3	2
Silhouette	≈ 0.53	≈ 0.26
ARI vs. reference (shared rows)	—	≈ 0.63

Deleting cases is not a weaker version of the same analysis —

it changes the partition.

The baseline collapses three tiers to two and halves the silhouette. How missing data is handled becomes part of the model's substantive output.

Tiers track clinical unit assignment



$$\chi^2 = 911 \quad p < 10^{-156}$$

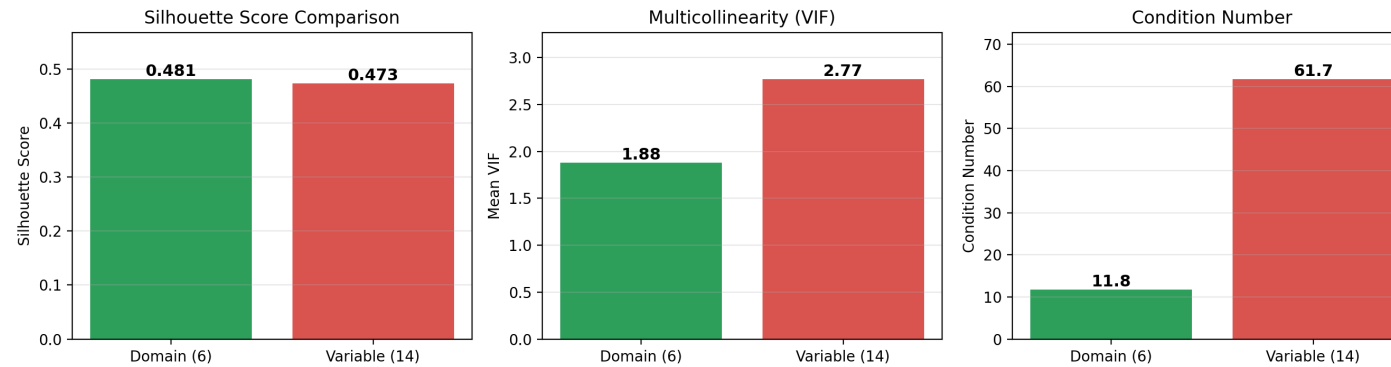
0.16

Cramér's V (pre-registered threshold 0.10)

- Cochran–Mantel–Haenszel: holds after adjusting for diagnosis.
- Severity tier adds information beyond the diagnostic label.
- Informs rehabilitation routing — it does not dictate placement.

Domain aggregation is more stable

Domain-Level vs Variable-Level Feature Comparison



Silhouette

~~0.473~~ → **0.481**

variables → domains

Mean VIF

~~2.77~~ → **1.88**

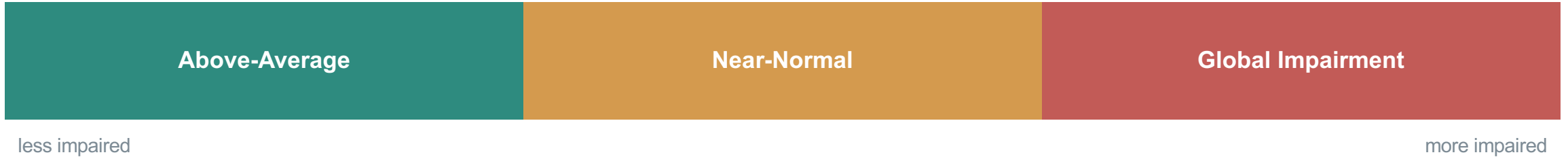
variables → domains

Condition number

~~61.7~~ → **11.8**

variables → domains

A continuum, not a catalogue



- Subtypes would imply qualitatively different cognitive profiles.
- A gradient implies differences lie mostly along one severity axis.
- The tiers are practical bands extracted from that continuum.
- Domain-specific dissociations may exist — but they don't dominate here.
- Poor scaling, redundant or non-cognitive features, or complete-case selection can fake subtype structure.

Clinical and analytic utility

Clinical users

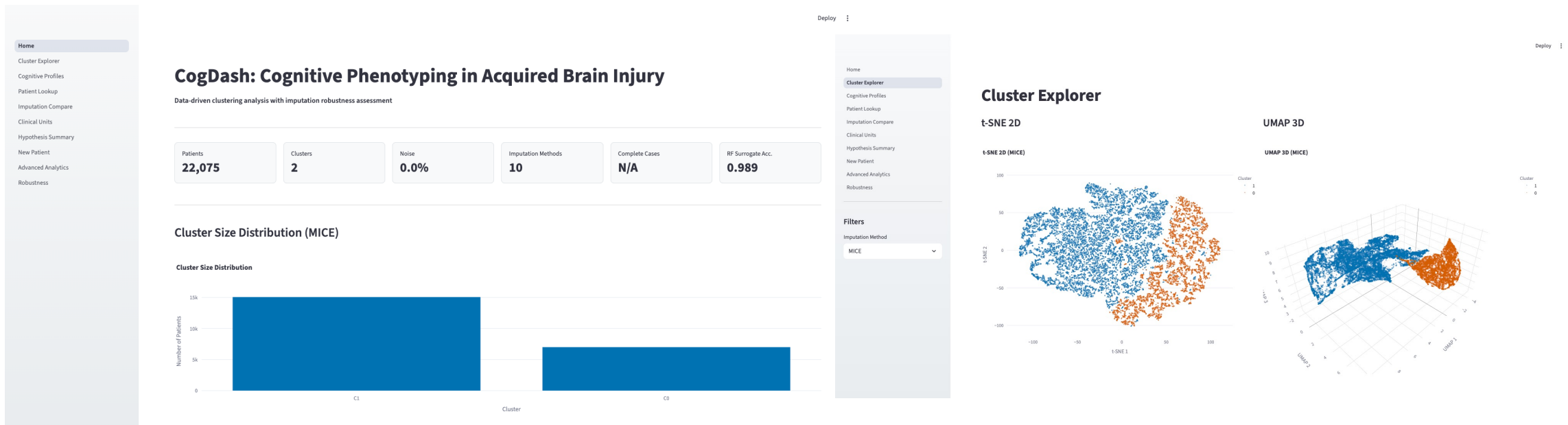
- Shared vocabulary for rehabilitation intensity.
- Confidence scores for borderline assignments.
- Information alongside diagnosis and judgement.

Analytic users

- Reproducible missingness-aware pipeline.
- Explicit, testable imputation decisions.
- Adaptable for linking cognition to outcomes.

Long-term contribution: make missing-data decisions explicit, then test their influence — rather than hiding them.

CogDash — a transparent decision-support tool



- Cluster explorer · patient lookup · new-patient classifier · hypothesis summaries.
- Built in Streamlit — auditable inputs and visible confidence.
- Supports collaboration; it does not automate clinical decisions.

Limitations

Single site

One rehabilitation centre — external replication is required.

Missing-data assumptions

MAR is a working assumption; MNAR cannot be excluded.

Embedding-based validation

UMAP-space silhouette may overstate separation.

Limited longitudinal modelling

Repeated assessments hint at recovery, but no full trajectory model.

No functional outcomes

Return-to-work, length-of-stay and independence were unavailable.

Pipeline sensitivity

Aggregation + density clustering may favour a general severity axis.

Three conclusions

1 Empirical

Routine post-ABI neuropsychological data carry a strong cognitive-severity gradient — ordered tiers, not discrete phenotypes.

2 Methodological

Imputation is a substantive modelling decision that can alter the recovered structure.

3 Practical

Domain-level severity tiers are interpretable and numerically stable — preferable to raw-variable clustering.

THANK YOU

Questions & discussion

A clinically useful severity continuum after acquired brain injury — more conservative than claiming discrete cognitive subtypes.

Zoltan Kunos · Supervisor: Dr. Alejandro García-Rudolph
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APPENDIX

Backup slides for examiner questions

Roadmap

- 1 Clinical problem and research gap
- 2 Cohort, missingness, and methods
- 3 Four pre-registered hypothesis tests
- 4 Interpretation and practical utility
- 5 Limitations and conclusions

Imputation hyperparameters

Method	Hyperparameters
Mean	Single-pass arithmetic mean per variable.
EM	Multivariate normal model; convergence 1e-6; max 500 iterations.
MICE	Linear regression per variable; 10 iterations; single imputed dataset for clustering.
KNN	k = 5; inverse-distance weighting; Euclidean distance.
MissForest	100 trees per forest; up to 10 iterations; variables imputed by increasing missingness.
PMM	5 donors; 10 iterations; implemented within the MICE framework.
SoftImpute	Regularisation chosen by CV over 20 log-spaced values (1e-3 to 1e2); convergence 1e-5; max 200 iterations.
NMF	Rank r = 5 selected by grid search over {2, 3, 5, 7, 10}; multiplicative updates; max 500 iterations.
DAE	15-128-64-32 autoencoder; ReLU + batch norm; dropout 0.5/0.3; 20% pattern-aware masking; MSE on observed entries; Adam (lr 1e-3); early stopping; 5 runs averaged.
VAE	Same layer dimensions as DAE; latent dim 32; loss = MSE + β -weighted KL (cyclical annealing, $\beta_{\max}$ 0.1); 20 latent samples averaged.

H2 fixed-reference design

- Fit the reference scaler on MICE-imputed data.
- Fit one UMAP embedding from the MICE reference.
- Define the reference HDBSCAN partition.
- Transform each alternative imputed dataset through the frozen pipeline.
- Compare predicted labels using pairwise ARI.

Tests

Robustness of point assignments to imputed values.

Does not test

Full invariance of independently discovered structures.

HDBSCAN logic

- Estimate local density via core distance.
- Construct mutual-reachability distances.
- Build a minimum spanning tree.
- Extract a hierarchy of density-connected components.
- Select persistent clusters by stability.
- Label points outside persistent regions as noise.

Unlike DBSCAN, HDBSCAN does not reduce to a single fixed- ϵ DBSCAN solution — it integrates across density levels.

Silhouette caveats

- UMAP is designed to emphasise neighbourhood structure.
- Silhouette measured in embedded space can overstate separation.
- Hyperparameter and seed selection make the chosen estimate optimistic.
- Seed stability, PCA and alternative representations provide supporting evidence.

Interpret silhouette as an internal pipeline-quality metric — not proof of natural categories.

Missing-not-at-random sensitivity

- MAR was the principal working assumption.
- MNAR mechanisms cannot be ruled out from observed values alone.
- Extreme-MNAR analyses suggest Global Impairment is comparatively robust.
- Boundaries between the less-impaired tiers are more sensitive.

Clinical interpretation boundary

The severity tier DOES

- supplement diagnosis;
- support rehabilitation discussion;
- carry a confidence estimate;
- assist outcome modelling.

The severity tier does NOT

- diagnose a patient;
- prescribe treatment;
- determine clinical placement;
- replace professional judgement.

Future work

- External replication across rehabilitation centres.
- Direct longitudinal modelling of recovery trajectories.
- Validation against return-to-work, length-of-stay and independence.
- Alternative model-based phenotyping (e.g. latent profile analysis).
- Prospective evaluation of CogDash in clinical workflows.
- Greater sensitivity to domain-specific dissociations.

How to read the metrics

Clustering quality & robustness

Silhouette

Cohesion vs. separation — how well a point fits its own tier vs. the nearest other.

-1 to 1; higher = cleaner. Pre-reg > 0.40 · observed ≈ 0.53

Adjusted Rand Index (ARI)

Agreement between two partitions, corrected for chance.

0 = chance, 1 = identical. Pre-reg > 0.50 · observed ≈ 0.71

Normalised Mutual Information

Shared information between two partitions, scaled 0–1.

0 = independent, 1 = identical · corroborates ARI

Consensus confidence

Share of assessments given a stable label across imputations.

> 86% high-confidence across methods

Association & conditioning

χ^2 statistic

Evidence that tier and clinical unit are associated — significance, not effect size; grows with N.

$\chi^2 = 911 \cdot p < 10^{-156}$

Cramér's V

Effect size of that association, scaled 0–1.

0.1 small · 0.3 medium · 0.5 large. Pre-reg > 0.10 · observed 0.16

VIF (variance inflation factor)

Per-feature redundancy from collinearity.

1 = none; > 5–10 concerning. 2.80 → 1.90

Condition number

Global conditioning of the feature set ($\lambda_{\max} / \lambda_{\min}$ of the correlation matrix).

< 10 fine · 10–30 moderate · > 30 strong. 62 → 12